TACTILE VIRTUALIZATION: RENDERING FABRIC TEXTURE AS SMARTPHONE VIBRATION

A PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

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This is to certify that the Major Project Phase-I titled **"TACTILE VIRTUALIZATION: RENDERING FABRIC TEXTURE AS SMARTPHONE VIBRATION"** is a record of bona fide work submitted by:

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The report is original, free from plagiarism, and complies with the Institute guidelines, including those related to academic ethics and AI-generated content.

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DECLARATION

We, the undersigned, declare that the Major Project Phase-I titled **"TACTILE VIRTUALIZATION: RENDERING FABRIC TEXTURE AS SMARTPHONE VIBRATION"** has been carried out by us in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech.)** in **COMPUTER SCIENCE AND ENGINEERING** of **Jawaharlal Nehru Technological University Hyderabad (JNTUH)**, under **VNRVJIET B.Tech. R22 Academic Regulations**, at **Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering & Technology, Hyderabad**.

We further declare that this work is original, has not been submitted previously for any degree or similar title, and complies with the Institute's plagiarism and academic ethics guidelines, including AI-generated content. The contributions of all team members are clearly indicated in the report.

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ABSTRACT

Online shoppers cannot touch the fabric of a garment before buying it, so they cannot judge how it feels. This project asks how much of a fabric's feel can be carried from a product photograph to the vibration motor of an ordinary Android smartphone, using software and public data only. The work is organised into five objectives: a haptic descriptor database with a pre-registered separability gate, visual fabric classification with a measurement of the gap between laboratory scans and product photographs, image-to-vibration association, a web rendering interface, and a perceptual user study. The literature shows that visual-to-tactile mapping, contrastive visual-tactile embeddings and velocity-conditioned rendering are already well studied. What remains open is whether models trained on laboratory surface scans transfer to real product photography, whether vibration descriptors stay stable across sliding speed, and what survives a phone's on/off vibration channel. Phase I used the Cluster Haptic Texture Dataset (118 materials, 16 of them cloth). The four-number vibration descriptor failed the pre-registered gate (silhouette −0.164, one effective class of 16), while a convolutional network on the spectrogram identified the fabric 37.8% of the time against 6.25% chance. A photo-trained material classifier recognised 75.6% of photographed fabric but only 3.3% of the dataset's fabric scans. After the vibration shared by every surface is removed, cloth alone shifts upward with sliding speed, which replicated on 2,240 held-out recordings. All results are saved, re-run from committed code and covered by 459 automated tests. Phase II will complete the product-photo transfer set, run the user study once ethics clearance is obtained, and submit the findings for publication.

Keywords: Haptic rendering, Vibrotactile feedback, Fabric texture, Domain transfer, E-commerce

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NOMENCLATURE

Symbol	Meaning
API	Application Programming Interface
CLIP	Contrastive Language–Image Pre-training (image–text model)
CNN	Convolutional Neural Network
CV	Coefficient of Variation
DTD	Describable Textures Dataset
ERM / LRA	Eccentric Rotating Mass / Linear Resonant Actuator (phone motors)
F1	Harmonic mean of precision and recall
FFT	Fast Fourier Transform
GPU	Graphics Processing Unit
ICC	Intraclass Correlation Coefficient
MAE	Mean Absolute Error
MINC	Materials in Context Database (MINC-2500 subset)
RMS	Root Mean Square
UML	Unified Modeling Language
f	Spectral centroid of acceleration (Hz)
v	Sliding velocity of the probe (mm/s)
λ	Spatial wavelength of a surface texture (mm)
ω^2	Omega-squared, an unbiased effect size for ANOVA
p	Probability value of a statistical test

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CHAPTER 1
INTRODUCTION

1.1 Background of the Problem

When a shopper buys a garment online, they judge its fabric from a photograph and a short text description. Touch, which is how people normally judge softness, roughness and drape, is missing. Haptics research has shown that much of what is felt when a finger or a tool slides over a textured surface is carried by small vibrations [1], [2]. Nearly every smartphone already contains a vibration motor, and Android browsers can drive it from a web page through the W3C Vibration API [3]. This suggests a software-only path to giving a shopper a rough sense of a fabric: map the product photograph to a vibration pattern and play it on the phone.

Doing this needs three things that do not yet exist together: recordings of how real fabrics vibrate when touched, a way to recognise the fabric from an ordinary product photograph, and a way to turn a recorded vibration into a pattern that a phone motor can play. Public datasets now provide the first. The Cluster Haptic Texture Dataset [4] records contact sound, probe acceleration, force and position for 118 materials, including 16 fabrics, at five sliding speeds, eight directions and two forces.

1.2 Need and Relevance

Experiments on mobile shopping show that adding vibrotactile feedback raises shoppers' anticipated satisfaction with a product [5], and that the value of haptic feedback depends on how much a shopper relies on touch [6]. A feature that lets a shopper feel even a coarse difference between, for example, corduroy and silk would add information that photographs cannot give. Because the method uses only a web page and the phone's existing motor, it needs no extra hardware and could reach any Android user. The work is also relevant to haptics research, because it measures how far laboratory haptic data can be reused for consumer devices.

1.3 Problem Statement

Given a photograph of a garment, produce a vibration pattern that a standard Android smartphone can play and that conveys the texture of the garment's fabric, using only public datasets and no custom hardware. Measure, at each stage, how much fabric-

specific information survives: from the surface to the recorded vibration, from the laboratory scan to the product photograph, and from the recorded vibration to the phone's on/off motor.

1.4 Aim and Objectives

The aim is to build and evaluate a pipeline that renders fabric texture as smartphone vibration from a product image, and to report honestly where information is lost. The objectives are listed below, and Table 1.1 gives how each one is measured and where it stands at the end of Phase I.

- O1 – Build a haptic descriptor database for the 16 cloth materials (frequency, intensity, pattern and duration) and test, with thresholds fixed in advance, whether the materials can be told apart.
- O2 – Train ImageNet-pretrained ResNet-50 classifiers for material (MINC-2500) and texture (DTD), and measure the accuracy gap between the dataset's 300 dpi scans and real product photographs.
- O3 – Compare two image-to-vibration association strategies (class-pair lookup and embedding similarity) on held-out materials, and test whether descriptors are invariant to sliding velocity.
- O4 – Build a web interface that goes from image to classifier to descriptor to waveform, spectrum and phone vibration, with a two-fabric comparison view.
- O5 – Validate the result perceptually with a user study (40 participants recruited, at least 30 usable, Android only), including mismatched control pairs.

Table 1.1 Objectives, how each is measured, and Phase I status

Objective	Measured by	Status at end of Phase I
O1 Haptic database	Coverage; ANOVA effect size; silhouette ≥ 0.30 and ≥ 8 of 16 materials separable (fixed in advance)	Done; gate failed (silhouette −0.164)
O2 Visual classification	Top-1 accuracy and macro F1; accuracy on scans vs ≥ 50 labelled product photos	Models trained (80.3%, 80.7%); photo set pending
O3 Association and velocity	Top-1/top-3 retrieval; descriptor error vs the mean; variation across 20–60 mm/s	Done; no strategy beats the mean; cloth-specific speed shift
O4 Interface	End-to-end latency; rendering on ≥ 3 Android models	Working prototype; device tests pending
O5 User study	Ratings against mismatched controls; ≥ 30 usable participants	Instrument ready; awaiting ethics clearance

1.5 Scope of Work

The project is software only; no hardware is built or bought. It uses public datasets only: the Cluster Haptic Texture Dataset [4], MINC-2500 [7], the Describable Textures Dataset (DTD) [8] and DeepFashion-MultiModal [9]. Rendering targets Android phones, because iOS browsers do not implement the Vibration API. The study is limited to the 16 fabrics in the Cluster dataset, with the other 102 materials used as a reference for what all surfaces share. Patent protection is out of scope, since software as such is excluded under Section 3(k) of the Indian Patents Act; software copyright registration is the appropriate route if needed.

1.6 Engineering, Environmental, Societal and Cost Considerations

Complexity: complex. The problem combines signal processing, deep learning on images, cross-domain transfer, statistics with pre-registered hypotheses, web development and human-subject evaluation.

Cost: low. All data is public, all software is open source, and model training uses free Kaggle GPU time (about 30 hours per week is available; 3.9 hours have been used so far). The user study needs only participants' own Android phones.

Environment and sustainability: if shoppers can judge fabric better before buying, fewer garments may be returned, which would reduce return shipping and waste. This is a possible benefit, not something this project measures. Compute use is kept small by running everything that does not need a GPU on a CPU.

Society: the feature could help shoppers who rely heavily on touch to judge clothing. The user study will follow the institute's ethics process and collect no identifying data beyond the phone model.

1.7 Organization of the Report

Chapter 2 reviews existing work and identifies the gaps. Chapter 3 describes the requirements, the approaches considered, the chosen pipeline and its design in UML. Chapter 4, the main chapter, reports the work completed in Phase I and its results. Chapter 5 lists the expected outcomes and the plan for Phase II. The appendices give each member's contribution and the project's relevance to the Sustainable Development Goals.

CHAPTER 2
LITERATURE REVIEW

This review builds on the team's survey of tactile virtualization, which classifies the field into measurement-driven and inference-driven approaches and compares consumer rendering technologies.

2.1 Overview of Existing Work

Why touch matters in online shopping. Li et al. found in several experiments that adding vibrotactile feedback to mobile shopping raised shoppers' anticipated satisfaction with a product, though not their purchase confidence [5]. Racat et al. show that the effect of haptic feedback in augmented-reality shopping apps depends on the shopper's need for touch [6], so haptic feedback cannot be treated as uniformly helpful.

Texture as vibration. Sliding over a fine texture produces vibrations whose content depends on both the surface and the scanning speed [2], and data-driven rendering reproduces texture by recording and replaying such vibrations [1].

Measurement-driven approaches. These record a physical surface directly. The Cluster Haptic Texture Dataset uses a 3-axis machine with a Shore A15 urethane rubber tip, sized to the contact area of an index finger, to slide over 118 surfaces at five speeds (20–60 mm/s), eight directions and two normal forces (0.5 and 1.0 N). With two repetitions this gives 160 recordings per surface and 18,880 in total, each with contact sound, acceleration, force and position [4]. Cao et al. instead press a GelSight optical tactile sensor onto 50 fabrics without sliding, producing the FabricVST dataset [10]. Lee et al. predict perceived smoothness and roughness of synthetic fibres from standardised mechanical measurements [11]. Braune et al. test whether an ordinary smartphone can imitate the roughness and friction of a textile [12]. Measurement gives accurate descriptors but needs specialised equipment for every new surface.

Inference-driven approaches. These predict touch from non-contact inputs. Discriminative models classify into material or texture classes: Zandonati et al. adapt vision foundation models to tactile data from four different sensors [13], and Park et al. improve visual texture classification with a feature retention module [14]. Generative models synthesise tactile signals: Ujitoko and Ban generate vibrotactile signals from texture images or attributes with a conditional GAN [15]; Cao et al. generate tactile

features from images and semantic attributes to recognise fabrics never touched before, reporting 83.06% accuracy on unseen materials [10]; TexSenseGAN lets a user steer a GAN's vibration output with a slider [16]; and Sound2Hap converts sound to vibration with a CNN autoencoder trained from human ratings [17]. Heravi et al. render texture vibration in real time from a surface image and the user's action [18]. Shared embedding spaces such as UniTouch align touch with vision, language and sound [19]. Retrieval approaches look up stored signals by nearest-neighbour search; FAISS and Annoy have been benchmarked for this on ResNet-50 image features [20].

Rendering technologies. Smartphone linear resonant actuators are widespread but narrowband [12]. Mid-air ultrasound can render clothing textures without contact; Lim and Yap built a mock clothing store with an Ultrahaptics array and found that it improved shoppers' satisfaction [21]. Wearable haptic textiles offer much finer spatial resolution [22]. Only the first is in every consumer's pocket.

Related surveys. Earlier reviews cover AI prediction of fabric properties and handfeel [23], deep learning for textile manufacturing [24], and tactile sensing for robot texture perception [25]. None connects sensing, consumer rendering and human evaluation, which is the gap the team's survey addresses.

Visual recognition. MINC-2500 [7] and DTD [8] provide labelled material and texture photographs; ResNet-50 [26] pretrained on ImageNet [27] is the standard starting point, and CLIP [28] classifies images without task-specific training.

2.2 Comparison of Methods / Approaches

Table 2.1 summarises the studies reviewed, in order of year, with the method, the key finding and the limitation that matters for this project.

Table 2.1 Literature survey of related work

No.	Study	Method	Key finding	Limitation for this project
1	Bensmaïa and Hollins, 2003 [2]	Texture vibration study	Skin vibrations depend on surface and scanning speed	Not a rendering method
2	Culbertson et al., 2014 [1]	Data-driven texture models	Records and replays tool–surface vibrations	Needs recordings of every surface
3	Ujitoko and Ban, 2018 [15]	Conditional GAN	Vibration generated from texture images or attributes	Texture images, not product photos

No.	Study	Method	Key finding	Limitation for this project
4	Braune et al., 2021 [12]	Smartphone rendering	Tests whether a phone can imitate textile roughness and friction	Narrowband phone actuator
5	Zandonati et al., 2023 [13]	Vision models on tactile data	Learns across four tactile sensors	Needs a tactile sensor
6	Cao et al., 2024 [10]	Zero-shot visual–tactile generation	83.06% on fabrics never touched; FabricVST (50 fabrics)	Static GelSight press, not vibration
7	Heravi et al., 2024 [18]	Learning-based real-time rendering	Texture vibration rendered in real time	Dedicated haptic device
8	Yang et al., 2024 [19]	Multimodal embedding (UniTouch)	Aligns touch with vision, language and sound	Vision-based tactile sensors
9	Zhang et al., 2024 [16]	GAN with user slider (TexSenseGAN)	Users steer generated vibration	Needs tuning by each user
10	Li et al., 2024 [5]	Shopping experiments	Vibration raises anticipated satisfaction, not purchase confidence	Not fabric-specific
11	Lim and Yap, 2024 [21]	Mid-air ultrasound store	Improved shopper satisfaction	Specialist hardware
12	Rahman et al., 2024 [20]	FAISS and Annoy benchmark	Fast nearest-neighbour search on ResNet-50 features	Not haptic
13	Eguchi et al., 2026 [4]	Machine-controlled sliding dataset	118 surfaces at 5 speeds, 8 directions, 2 forces	Laboratory scans, not product photos
14	Racat et al., 2026 [6]	AR shopping experiment	Effect depends on the shopper's need for touch	Not fabric rendering
15	Li and Seifi, 2026 [17]	CNN autoencoder (Sound2Hap)	Sound converted to vibration, learned from ratings	Audio input, not images

Table 2.2 Rendering technologies for consumer use

Technology	Example	Spatial detail	Consumer availability
Phone linear resonant actuator	Braune et al. [12]	Whole device vibrates	In almost every phone
Mid-air ultrasound	Lim and Yap [21]	Focal points in mid-air	Specialist hardware
Breathable haptic textile	Yao et al. [22]	Multi-site array on the hand	Wearable prototype
Dedicated haptic device	Heravi et al. [18]	Single point	Research hardware

2.3 Critical Analysis of Literature and Inferences

- Visual-to-tactile generation, shared vision–touch embeddings and action-conditioned rendering are all established [15], [19], [18]; none of them is a new contribution here.
- Most rendering work uses dedicated hardware [18], [21], [22]. A phone motor is a far narrower channel [12], and how much texture survives it is rarely measured.
- Measurement-driven data is accurate but tied to laboratory rigs [4], [10]; inference-driven models depend on large labelled sets and are rarely tested on real product photography.
- Consumer benefit is real but uneven: vibration raises anticipated satisfaction [5], but its value depends on the shopper's need for touch [6], so user evaluation must be part of the design.
- Speed matters for vibration [2], [18], yet few datasets vary it; the Cluster dataset's five speeds make velocity effects testable [4].

2.4 Research Gaps Identified

Table 2.3 lists the three gaps this project addresses, how it addresses each, and what Phase I found.

Table 2.3 Research gaps and how this project addresses them

Gap	Description	How this project addresses it	Phase I result
G1 Scan to photo	Do models built on laboratory scans transfer to e-commerce photos? (headline question)	Accuracy on scans vs labelled product photos (O2)	Gap runs both ways: 0.675 vs 0.287; CLIP 0.635 vs 0.460
G2 Velocity	Are fabric descriptors stable across sliding speed?	Descriptor variation across 5 speeds (O3)	Centroid ignores speed; cloth shifts by about a third of v/λ
G3 Binary channel	What survives a motor that can only switch on and off?	10 ms on/off encoding and fidelity (O4)	9 of 80 envelopes playable; identity above chance

CHAPTER 3
PROPOSED DESIGN / SOLUTION APPROACH

3.1 Requirements and Constraints

Functional requirements:

- Accept a garment image from the user's phone or computer.
- Classify the fabric's material and texture from the image.
- Retrieve or predict a vibration descriptor for that fabric.
- Show the waveform and spectrum, and play the pattern on the phone's vibration motor.
- Compare two fabrics side by side, and log the device model and latency of each session.

Non-functional requirements and constraints: software only; public datasets only; about six months and four people; runs in an Android browser (iOS cannot vibrate from a web page); end-to-end latency must be short enough for interactive use; every result must be reproducible from saved code and data; training must fit within the free Kaggle GPU quota.

Table 3.1 links each main requirement to the design control that meets it and to how it is checked.

Table 3.1 Principal requirements and design controls

Requirement	Design control	How it is checked
Reproducibility	Every run saved with its configuration, metrics, Git commit and source hash	Quoted results re-run from committed code (Appendix D)
Honest evaluation	Thresholds and predictions written down and committed before data is analysed	Registration file in the repository
Generalisation	One sliding direction held out in every identification test	8-fold cross-validation
Phone playback	On/off patterns on a 10 ms tick through the Vibration API	Fidelity scores now; device tests in Phase II
Compute budget	CPU for everything except CNN training; resumable Kaggle jobs	3.9 GPU-hours used of about 30 per week
Correctness	Automated tests designed to fail on bad data	459 tests

3.2 Possible Approaches Considered

Table 3.2 Approaches considered and the decision taken

Approach	Decision	Reason
Improve a published generative model (e.g. a texture GAN)	Rejected	Depends on weights that may not be public, and would need to beat the original
Collect a new dataset (including phone-microphone capture)	Rejected	Team decided to use public data only
Build or buy haptic hardware	Rejected	Software-only scope agreed with the guide
Measure transfer and information loss with public data, phone rendering	Chosen	Feasible, testable, and addresses open questions

3.3 Proposed Approach / Methodology

The pipeline has five stages, each matching one objective. Fig. 3.1 shows the system architecture: the public datasets, the modules for each objective, and the run log that stores every result.

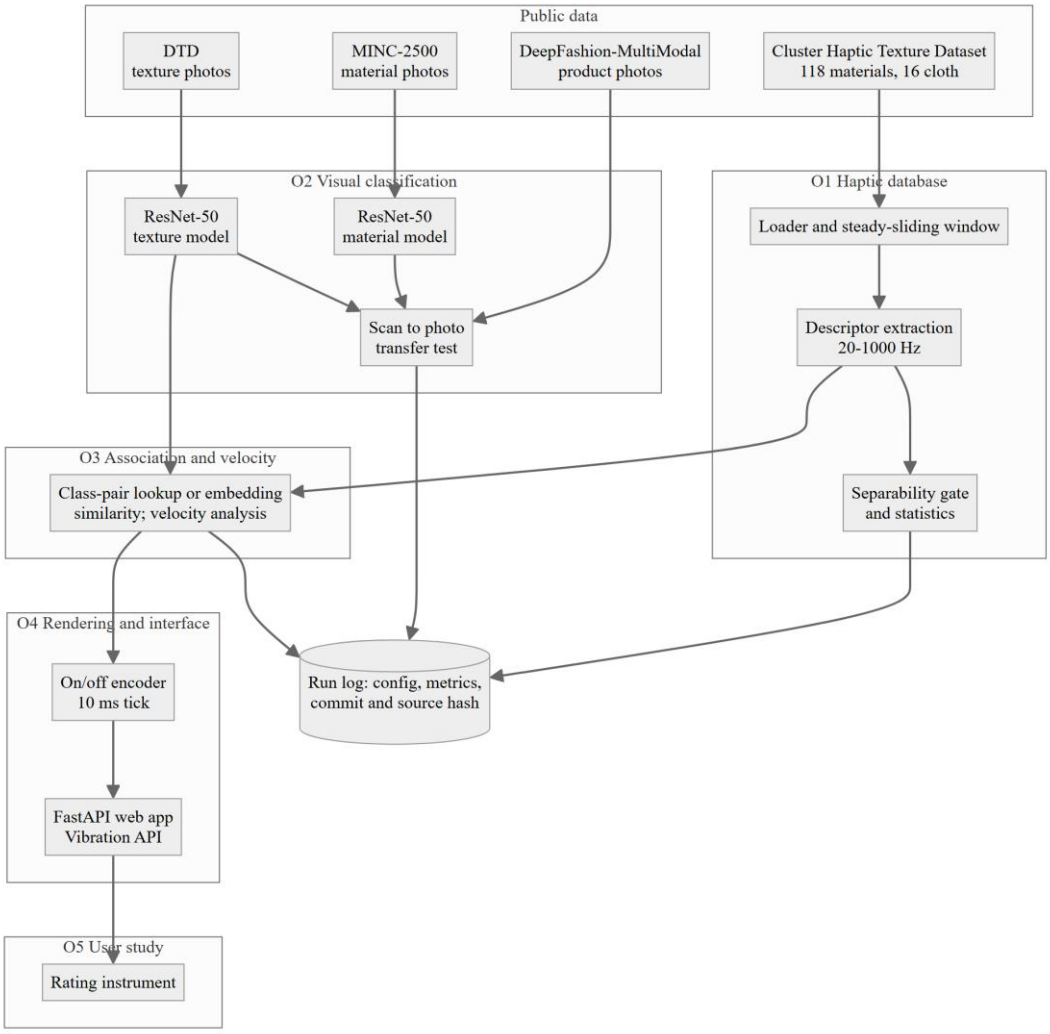


Fig. 3.1 System architecture

- Haptic database (O1): load the cloth recordings, band-pass the acceleration to 20–1000 Hz, and compute four descriptors – spectral centroid (frequency), RMS (intensity), normalised spectral entropy (pattern) and dominant envelope period (duration). Recordings whose energy falls below the band are flagged and excluded.
- Visual classification (O2): fine-tune two ResNet-50 models [26] pretrained on ImageNet [27], one on MINC-2500 [7] for material and one on DTD [8] for texture, restricted to 12 textile classes.
- Association (O3): map an image to a descriptor either by class-pair lookup or by cosine similarity of image embeddings.
- Rendering (O4): convert the descriptor into an on/off pattern with a 10 ms tick and play it with the Vibration API [3] through a FastAPI web service.
- Perceptual validation (O5): participants rate the match between images and vibrations, including mismatched controls.

Fig. 3.2 shows the same pipeline as an activity diagram, from upload to playback.

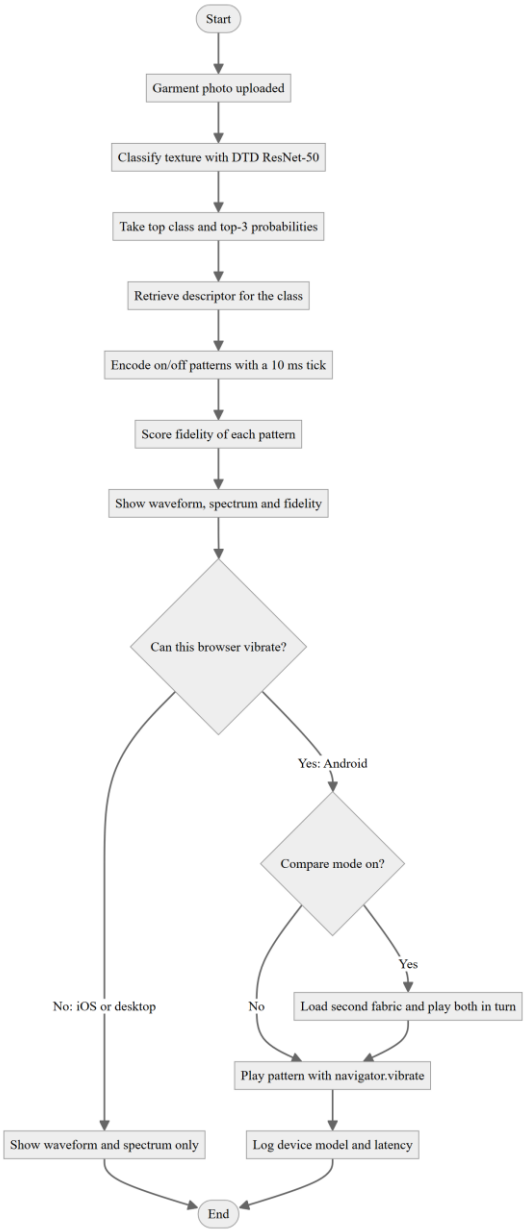


Fig. 3.2 Activity diagram of the image-to-vibration pipeline

The diagrams in this section are generated from the same sources that StarUML imports (report/uml/). Use case view. The shopper uploads an image, views the predicted fabric and vibration, plays it, and compares two fabrics. The researcher runs the descriptor gate, trains the classifiers and evaluates transfer (Fig. 3.3).



Fig. 3.3 Use case diagram

Structure. The implementation is organised into Python packages: haptic (descriptors, loader, statistics, rendering), vision (classifiers and transfer), webapp (the interface) and runlog (saved, versioned results). Fig. 3.4 shows the main classes.

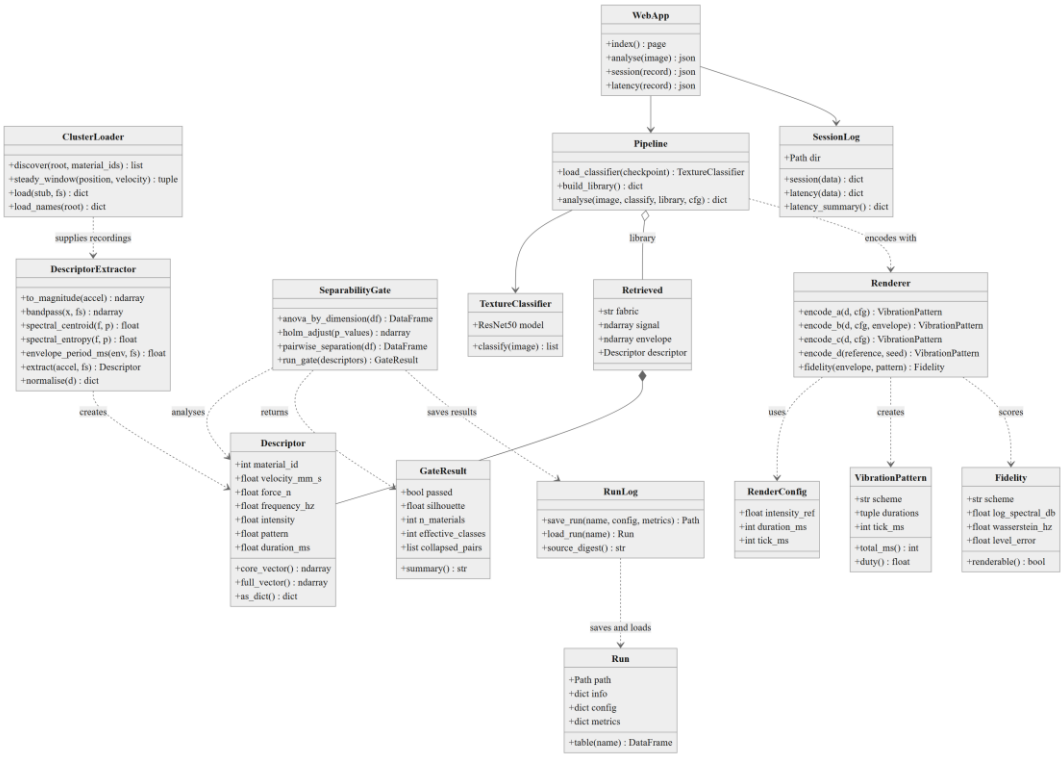


Fig. 3.4 Class diagram of the main modules

Interaction. Fig. 3.5 shows the sequence of calls when a shopper asks to feel a fabric.

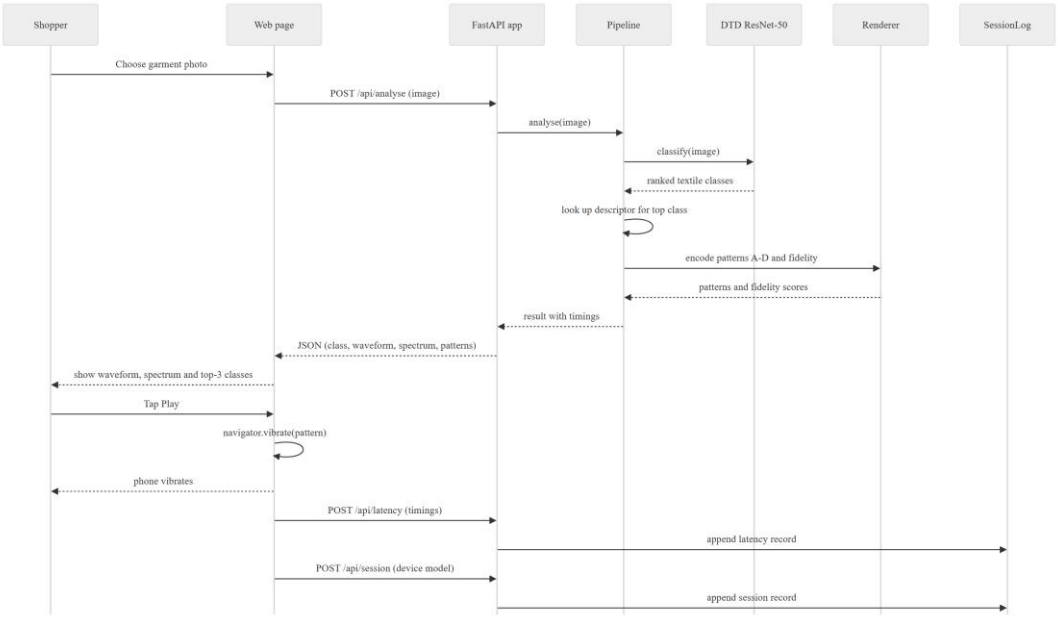


Fig. 3.5 Sequence diagram for “feel this fabric”

States. Fig. 3.6 shows the states of a rendering session in the web interface.

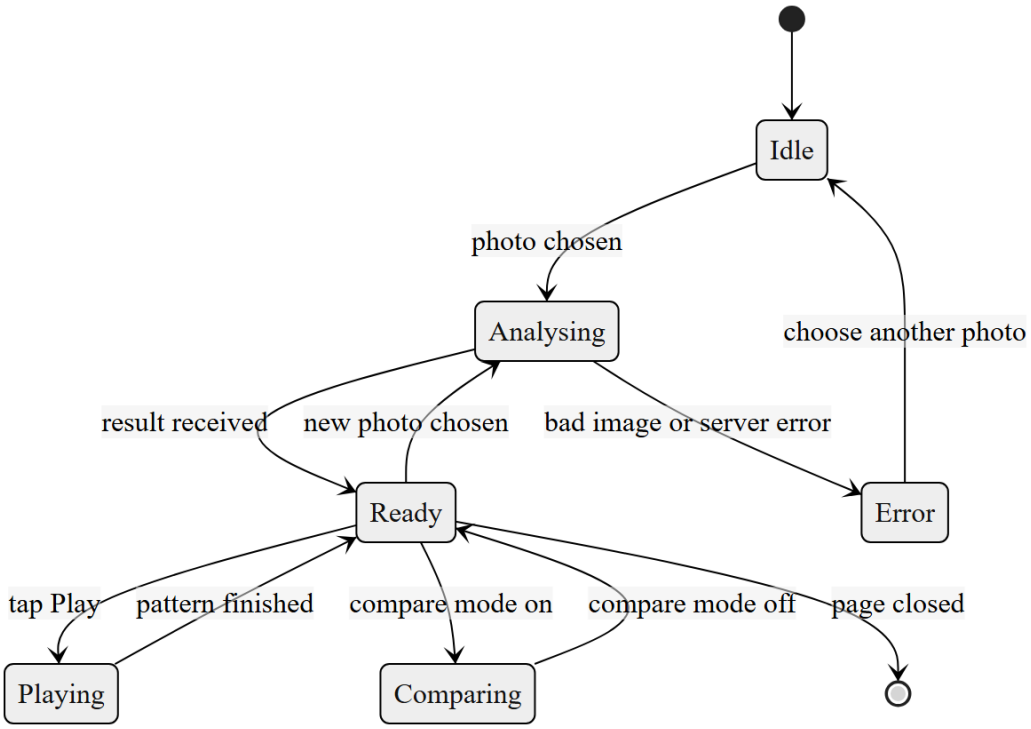


Fig. 3.6 State diagram of a rendering session

3.4 Technical Justification and Supporting Analysis

- Four descriptors match the perceptual dimensions most often used for vibrotactile texture (frequency, intensity, regularity and rhythm) and fit what a phone can render.
- Normalised spectral entropy replaced spectral flatness, because flatness returned about zero for every material on these narrow-band signals.
- Omega-squared replaced eta-squared as the effect size, because eta-squared is biased upward with small samples. Holm correction [29] controls false positives across the four dimensions.
- The envelope is decimated to about 500 Hz before its period is estimated; at the raw 6 kHz rate the frequency resolution was too coarse and periods came out three times too long.
- Results are validated with sliding direction held out, so a model cannot succeed by memorising one stroke.

3.5 Assumptions and Benchmarks

- Assumption: the probe's acceleration in the Cluster dataset reflects what a finger would feel well enough to compare fabrics. Phase I tested part of this (Chapter 4).

- Benchmark for O1, fixed before any real data was analysed: pooled silhouette of at least 0.30 and at least 8 of the 16 materials separable. If fewer than 8 separate, the class list is coarsened and the grouping reported.
- Benchmarks for identification: chance is 1/16 = 6.25% for the 16 fabrics and 1/118 = 0.85% for all materials.
- Benchmark for O2: accuracy on the dataset's scans versus at least 50 product photographs labelled by two people.

3.6 Tools / Techniques Proposed

Table 3.3 Tools and techniques

Category	Tools
Language and libraries	Python 3.13, NumPy, SciPy, pandas, scikit-learn
Deep learning	PyTorch, torchvision (ResNet-50), Hugging Face transformers (CLIP)
Compute	Local CPU; Kaggle notebooks with an NVIDIA T4 GPU for training
Web interface	FastAPI, HTML/JavaScript, W3C Vibration API
Testing and versioning	pytest (459 tests), Git and GitHub, run logs with source hashes
Design	StarUML (UML diagrams), Matplotlib (figures)

3.7 Feasibility and Sustainability Considerations

Technical feasibility is shown by the Phase I implementation, which runs end to end on public data. Economic feasibility follows from the zero hardware and data cost. Operational feasibility is limited by two factors: iOS phones cannot be used, and the user study depends on ethics clearance, whose date the team does not control. The plan therefore keeps the publishable results independent of the study (Chapter 5). The code is modular, tested and versioned, so it can be maintained and extended after the project.

CHAPTER 4
PRELIMINARY DESIGN, ANALYSIS AND PHASE–I PROGRESS

4.1 Phase I Activities and Progress

After the team's literature survey, the implementation work recorded in the project repository ran from 31 October to 8 November 2026. In that period the team set up the data pipeline for the Cluster Haptic Texture Dataset [4], fixed the success thresholds in writing before touching real data, built the descriptor database, trained the image classifiers, prototyped the web interface, and then ran six further rounds of analyses. Table 4.1 gives the status of each objective and Table 4.2 the order of work. Alongside the code, the team wrote a survey paper, "A Survey of Tactile Virtualization: Taxonomy, Analysis, Evaluation, and Future Directions", which classifies the field into measurement-driven and inference-driven work and compares consumer rendering technologies. Its 17 references were checked against publisher and arXiv records; the text is being revised for submission. Chapter 2 is built on it.

Table 4.1 Status of each objective at the end of Phase I

Objective	Work completed	Status	Where reported
O1 Haptic database and gate	Descriptors for 2,493 cloth recordings; gate; variance analysis; group test; CNN baseline	Complete	Sec. 4.4.1
O2 Visual classification	MINC-2500 model; 10 DTD folds; tested on 1,600 scan crops; first product-photo transfer test	Mostly complete; labelled photo set pending	Sec. 4.4.2
O3 Association and velocity	Two strategies on held-out materials; velocity analysis on all 118 materials; replication	Complete for Phase I	Sec. 4.4.3
O4 Interface	FastAPI web service with waveform, spectrum, four vibration schemes, comparison view and logging	Prototype; retrieval still a placeholder	Sec. 4.3, 4.4.4
O5 User study	Rating instrument, reliability and analysis code; labelling tool	Waiting for ethics clearance	Sec. 5.3
Survey paper	Literature survey written; 17 references checked	Under revision	Chapter 2

Table 4.2 Phase I timeline

Period (2026)	Work done
Before 31 Oct	Literature survey; survey paper drafted (A Survey of Tactile Virtualization: Taxonomy, Analysis, Evaluation, and Future Directions, 17 references), now under revision
31 Oct	Objective 1 descriptor extraction and gate; ResNet-50 training on MINC-2500 and DTD; gate thresholds registered

Period (2026)	Work done
1–4 Nov	All 10 DTD folds; cloth group assignment registered; O1 gate, O2 scan test, O3 and O4 on real data
5–6 Nov	Velocity analysis on all 118 materials; held-out replication; CNN baseline on Kaggle GPU
7 Nov	Rounds 2–5: 15 registered analyses (A–O) on 18,614 recordings of all 118 materials
7–8 Nov	Round 6 (whitened CNN, GPU); provenance corrections; all quoted results re-run from committed code

4.2 Preliminary Design / Model / Calculations / Simulation

Data. The 16 cloth materials of the Cluster dataset (IDs 102–117) are listed in Table 4.3 with the number of usable recordings of each. Every material was recorded at 5 speeds, 8 directions and 2 forces, twice (160 recordings); recordings with less than 0.5 s of steady sliding or with energy below the analysis band were excluded, leaving 2,493 [4].

Table 4.3 The 16 cloth materials and their usable recordings

ID	Material	Recordings	ID	Material	Recordings
102	Velvet	145	110	Organdy	157
103	Corduroy	153	111	Mesh	158
104	Cotton	152	112	Wool	157
105	Cotton broadcloth	157	113	Nylon	154
106	Cotton gauze	159	114	Polyester	156
107	Chiffon	158	115	Silk	157
108	Lace	158	116	Linen	156
109	Georgette	157	117	Cotton linen	159

Descriptor definitions. Acceleration is band-passed to 20–1000 Hz, and four numbers are computed for each recording (Table 4.4). Recordings whose energy falls below the band, which happens for coarse fabrics at low speed, are flagged and excluded rather than averaged in.

Table 4.4 Haptic descriptor definitions

Descriptor	Definition
frequency_hz	Spectral centroid of the band-passed acceleration
intensity	RMS acceleration in the band
pattern	Normalised spectral entropy, 0–1 (low = periodic, high = random)
duration_ms	Dominant period of the envelope modulation (envelope decimated to about 500 Hz first)
wavelength_mm	v / f (velocity-normalised form)
roughness_index	intensity / (force × √(v/40))

The four descriptors and the derived quantities are defined by Eqs. (4.1)–(4.5), where P_k is the Welch power at frequency f_k inside 20–1000 Hz, x_b is the band-passed acceleration of N samples, v the sliding speed (mm/s) and F the normal force (N).

$$f_c = \sum_k f_k P_k / \sum_k P_k \tag{4.1}$$

$$I = \sqrt{((1/N) \sum_n x_b[n]^2)} \tag{4.2}$$

$$H = -\sum_k q_k \ln q_k / \ln K, \quad q_k = P_k / \sum_j P_j \tag{4.3}$$

$$T = 1000 / f_{env} \ ms, \quad f_{env} = peak\ of\ the\ envelope\ spectrum \tag{4.4}$$

$$\lambda = v / f_c, \quad R = I / (F \sqrt{(v / 40)}) \tag{4.5}$$

Separability and effect size use Eqs. (4.6) and (4.7), where a(i) is the mean distance from recording i to its own material and b(i) the mean distance to the nearest other material. Whitening, Eq. (4.8), subtracts from each feature its median over all 118 materials at the same speed, force and direction c.

$$\omega^2 = (SS_b - (k - 1) MS_w) / (SS_t + MS_w) \tag{4.6}$$

$$s(i) = (b(i) - a(i)) / max\{a(i), b(i)\} \tag{4.7}$$

$$\tilde{x}_m(c) = x_m(c) - median_{m'} x_{m'}(c), \quad c = (v, F, \theta) \tag{4.8}$$

Statistical design. Every confirmatory test was written into a registration file and committed to Git before it was run, with its prediction and threshold. Effects across materials use one-way ANOVA with omega-squared and Holm correction [29]. Separability uses the silhouette score and a pairwise distance matrix with a collapse threshold of 1.0 standard deviation. Identification is tested with logistic regression and with a small CNN on the spectrogram, always with one sliding direction held out (8 folds), so that a model cannot succeed by memorising one stroke. Comparisons between methods use one-sided Wilcoxon tests over folds.

Rendering model. A phone's vibration motor can only be switched on and off, and the Vibration API takes a list of on/off durations. The project fixes a 10 ms tick, which means the fastest pattern a phone can play repeats at 25 Hz. Four schemes turn a descriptor into an on/off pattern (A–C) and a random pattern of the same length serves as a control (D). A 25 ms motor lag is assumed when comparing what was intended with what a phone can play.

4.3 Preliminary Implementation or Proof of Concept

All code is in a single Git repository with 48 commits and 459 automated tests. The main parts are:

- A dataset loader that reads the Cluster files, cuts each recording to its steady sliding stretch using the position data, and fails clearly if the file layout is not as expected.
- Descriptor extraction, the separability gate and the statistics, with tests that are designed to fail on bad data.
- ResNet-50 [26] training scripts packaged for Kaggle, with resumable checkpoints; 3.9 GPU-hours used in total.
- A transfer evaluation for product photographs, and an offline labelling page so that two people can label photographs independently.
- A FastAPI web service and page: the user uploads an image, sees the predicted texture class with its top-3 probabilities, the waveform, spectrum and fidelity of four vibration schemes, plays the pattern on an Android phone, compares two fabrics, and the page logs device model and latency.
- A run log that saves every result with its configuration, Git commit and a hash of the source code, so every number in this report can be traced to the exact code that produced it.

Fig. 4.1 shows the interface after it has analysed one of its sample images: the predicted texture class with its top-3 probabilities, the descriptor, the waveform and its spectrum, with the band that a 10 ms tick can play shaded.



Fig. 4.1 The web interface after analysing a sample image (desktop browser)

Data design. Table 4.5 lists the main data stores. Raw data stays outside version control; everything derived from it is saved with enough information to regenerate it.

Table 4.5 Main data stores

Store	Contents	Format
Cluster mirror (local)	Cloth recordings (acceleration, sound, force, position) and 300 dpi scans	CSV, FLAC, JPEG; not in Git
Feature table	One row per recording: material, speed, force, direction, repeat, descriptors, 16 band powers, in-band flag	CSV, 18,614 rows
Run log (results/)	Per run: run.json (time, commit, source hash), config.json, metrics.json and tables	JSON and CSV

Store	Contents	Format
Models	ResNet-50 checkpoints: MINC-2500 fold 1, DTD folds 1–10	PyTorch .pt
Interface logs	Session records (device model) and latency records	JSON lines
Registration file	Predictions and thresholds, committed before each analysis	Markdown in Git

Limitation of the prototype: the step from texture class to vibration descriptor in the web interface is still a hand-made placeholder, and the page marks its output as synthetic until the real retrieval is connected in Phase II.

4.4 Preliminary Checks, Tests and Observations

4.4.1 Objective 1: can the four descriptors tell 16 fabrics apart?

Of 2,560 cloth recordings, 47 had less than 0.5 s of steady sliding and 20 fell below the analysis band, leaving 2,493. Table 4.6 shows the gate result; the thresholds were fixed before the data was used.

Table 4.6 O1 separability gate on the 16 cloth materials

Measure	Raw descriptors	Velocity-normalised	Threshold
Pooled silhouette	−0.164	−0.124	≥ 0.30
Collapsed pairs	120 of 120	120 of 120	—
Effective classes	1	1	≥ 8
Verdict	FAIL	FAIL	

Every descriptor differs significantly across materials (Holm $p < 10^{-15}$), but material explains only 8–15% of each one, and sliding direction and recording-to-recording variation explain most of the rest (Fig. 4.2). Because every pair of fabrics overlaps, coarsening the class list, the plan's fallback, cannot rescue the gate. The registered group test also failed: the seven woven fabrics do not cluster (Holm $p = 0.23$), and the pile group had too few members to test.

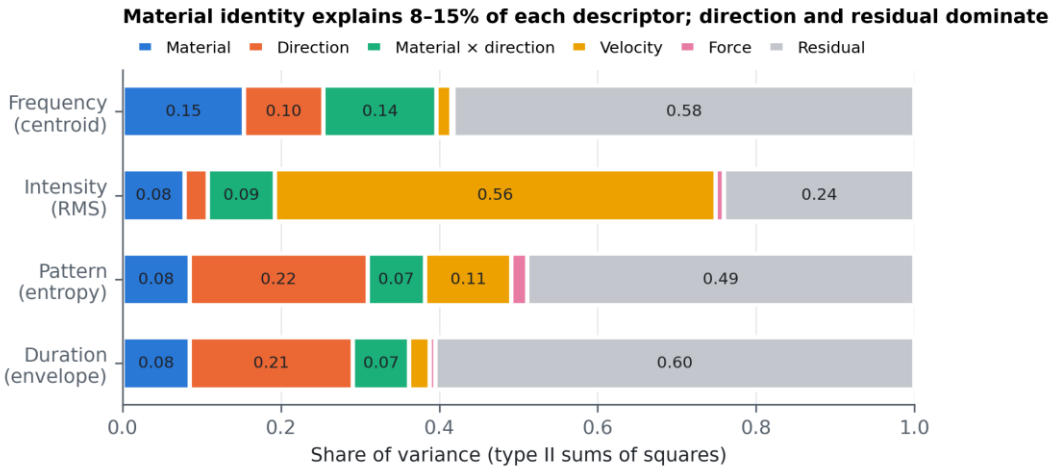


Fig. 4.2 Share of variance in each descriptor

To find out whether the problem is the descriptor or the signal, the same task was run with richer inputs (Fig. 4.3). With direction held out, the four descriptors identify the fabric 13.8% of the time, a 16-band spectrum 21.8%, and a CNN on the spectrogram 37.8%, against 6.25% chance. So the signal carries much more identity than four numbers keep. Two repeats of the same stroke agree closely (19 of 21 features have an intraclass correlation of at least 0.75), so the overlap is not measurement noise.

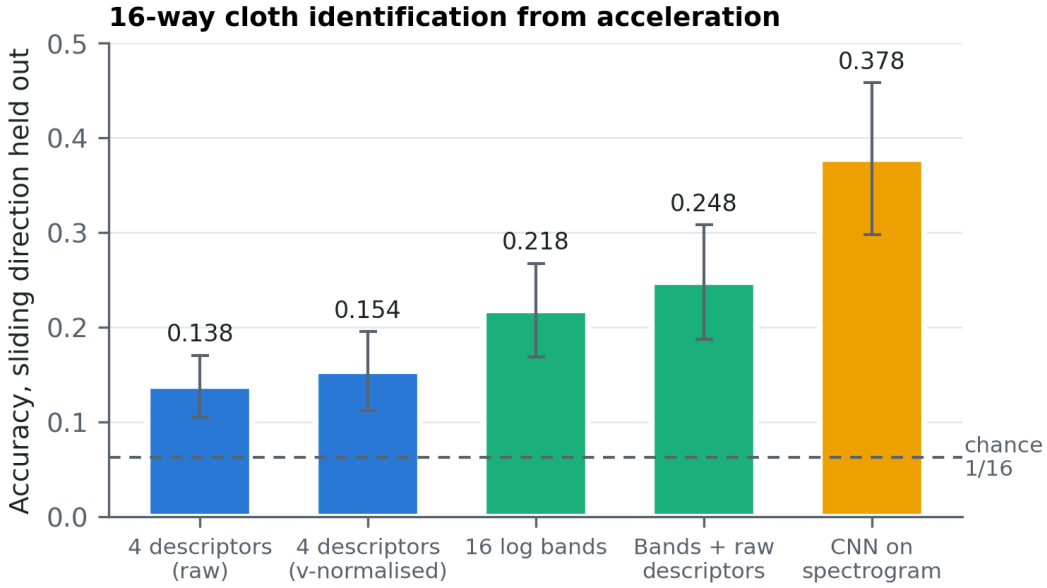


Fig. 4.3 Sixteen-way fabric identification from acceleration

4.4.2 Objective 2: image classifiers and the scan-to-photo gap

The MINC-2500 material model reaches 80.3% top-1 accuracy (macro F1 0.802) on 23 material classes [7]. The DTD texture models reach 69.2% on all 47 classes and 80.7% on the 12 textile classes (macro F1 0.806), mean over 10 folds with a spread of 1.9

points [8]. Fig. 4.4 shows the 12 textile classes over all 10 folds: braided (62%), fibrous (72%) and woven (73%) are the hardest; fibrous is most often mistaken for matted, and braided for woven.

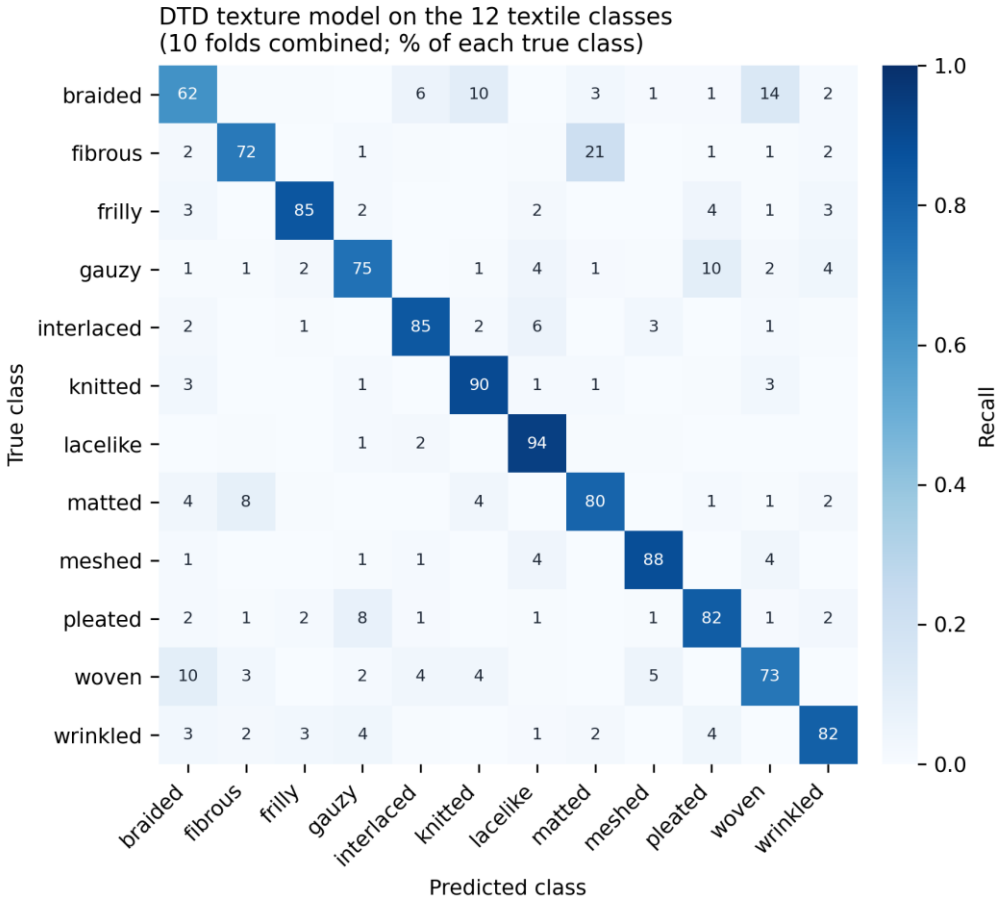


Fig. 4.4 Confusion matrix of the DTD texture classifier on the 12 textile classes

On the dataset's 300 dpi fabric scans the picture changes completely. The MINC model calls 75.6% of photographed fabric “fabric” but only 3.3% of the 1,600 scan crops, mostly “carpet” or “wallpaper” instead, and none of the 16 fabrics is called fabric by majority. Showing it a larger field of view does not help (Fig. 4.5).

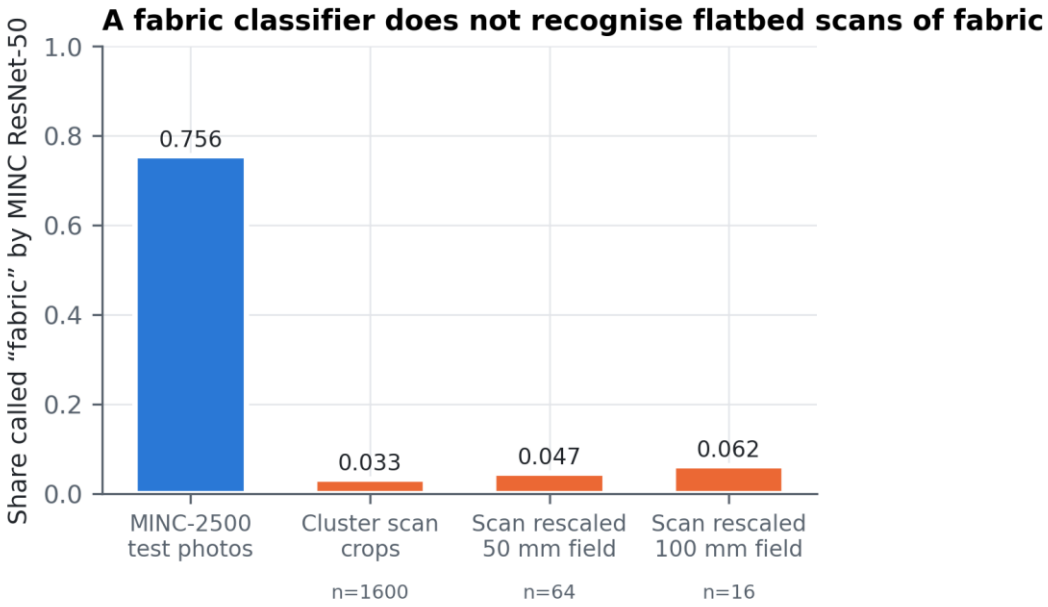


Fig. 4.5 A photo-trained classifier does not recognise flatbed scans of fabric

A first test in the other direction used DeepFashion-MultiModal product photographs [9] labelled cotton, chiffon or leather from their captions. A classifier trained on scans was perfect on scans (1.000) but scored 0.234 on the photographs, below the 0.333 chance level. A stricter re-test on unseen swatches gave 0.675 on scans against 0.287 on photographs (chance 0.5). CLIP [28], trained on web photographs, shows the mirror image: 0.635 on photographs but 0.460 on scans. The gap therefore runs both ways, and it is the headline result that Phase II will measure properly on a labelled set.

4.4.3 Objective 3: association and sliding velocity

Retrieval. Two strategies were compared for predicting a held-out fabric's descriptor from its image: class-pair lookup and cosine similarity of image features. Neither beat simply predicting the average descriptor, and class-pair lookup was significantly worse for intensity and duration (Holm $p = 0.026$). This follows from O1: when the descriptors barely differ between fabrics, choosing by appearance cannot beat the average. Across all 118 materials, looking similar predicts vibrating similarly only between categories ($r = +0.154$), not within them, and not among the 16 fabrics.

Velocity. On this rig the spectral centroid barely changes with sliding speed for any of the 118 materials: none is within 0.25 of the $f = v/\lambda$ prediction, and every category sits at 240–270 Hz. Dividing by speed therefore makes descriptors worse. After removing the spectrum shared by all surfaces, however, cloth alone shifts upward with speed, by

about a third of v/λ (median exponent 0.338, all 16 fabrics above zero; Fig. 4.6). This was found in exploratory analysis and then replicated on 2,240 held-out recordings.

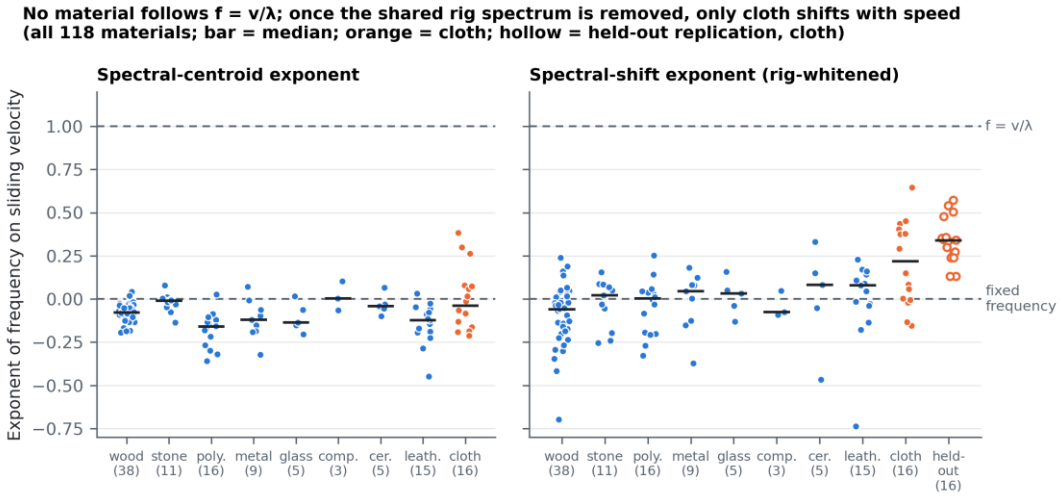


Fig. 4.6 Velocity exponents for all 118 materials

4.4.4 Objective 4: what survives the phone's on/off channel

Only 9 of 80 real vibration envelopes can be reproduced with a 10 ms tick, because most of their modulation is faster than 25 Hz. On those that can, the best scheme (B) differs from the intended envelope by 8.4 dB, but the random control (D) scores 7.5 dB, so this metric cannot tell them apart; that question needs people. Fabric identity does survive rendering partly: rendered patterns identify the fabric 9.9% of the time against 6.25% chance, while the intended patterns reach 11.8%.

4.4.5 Further registered analyses (rounds 2–6)

Sixteen more analyses (A–P) were registered and run on all 18,614 in-band recordings of the 118 materials. Table 4.7 lists every prediction and whether it held. Two results stand out. Contact sound carries far more identity than acceleration (72.0% against 37.8% for the 16 fabrics), but mostly above 1 kHz, which a phone motor cannot play. And removing the response that every surface shares ("whitening") improves identification for both a linear model and the CNN, reaching 43.5% for the 16 fabrics, the best result from vibration alone.

Table 4.7 Registered predictions in rounds 2–6 and their outcomes

Analysis	Result	Outcome
A. Looks predict feel (118 materials)	r = +0.154 overall; none within category or within cloth	Held
B. Sound vs acceleration, 16 fabrics	72.0% vs 37.8% (8/8 folds)	Held

Analysis	Result	Outcome
C. Speed correction k = 0.338 vs none	30.5% vs 27.9% at an unseen speed	Held
C. k = 0.338 vs full v/λ (k = 1)	30.5% vs 28.4%, p = 0.064	Not supported
E. Stroke direction structure	Perpendicular > opposite in 16/16; strongest in glass and metal (rig effect)	Held
F. Identity after on/off rendering	9.9% vs 6.25% chance	Held
F. Intended beats rendered	11.8% vs 9.9%	Held
H. Training on one force, testing on the other	16 fabrics 32.4% → 22.4%; 118 materials 12.6% → 6.7%	Held
G. Scan model on product photos	1.000 on scans vs 0.234 on photos	Held
I. Whitening, linear model	16 fabrics 21.8% → 30.6%; 118 materials 7.8% → 13.3%	Held
J. Strict scan vs photo; CLIP	0.675 vs 0.287; CLIP 0.635 on photos vs 0.460 on scans	Held
L. Category from image, then descriptor	Category 72.9% correct; only the centroid improves (1 of 4)	Partly held
M. Sound vs acceleration, 118 materials	22.5% vs 7.8% (8/8 folds)	Held
N. Whitening rescues the O1 gate	Silhouette worsened, −0.07 → −0.15	Failed
O. Speed dependence removed by whitening	73% shared by all surfaces; cloth keeps the most	Held
P. Whitening, CNN	16 fabrics 38.1% → 43.5%; 118 materials 20.2% → 31.4%	Held

Fig. 4.7 brings the identification results together. For both the 16 fabrics and all 118 materials, accuracy rises steadily from the four descriptors to the whitened CNN, and contact sound sits above vibration for the fabrics.

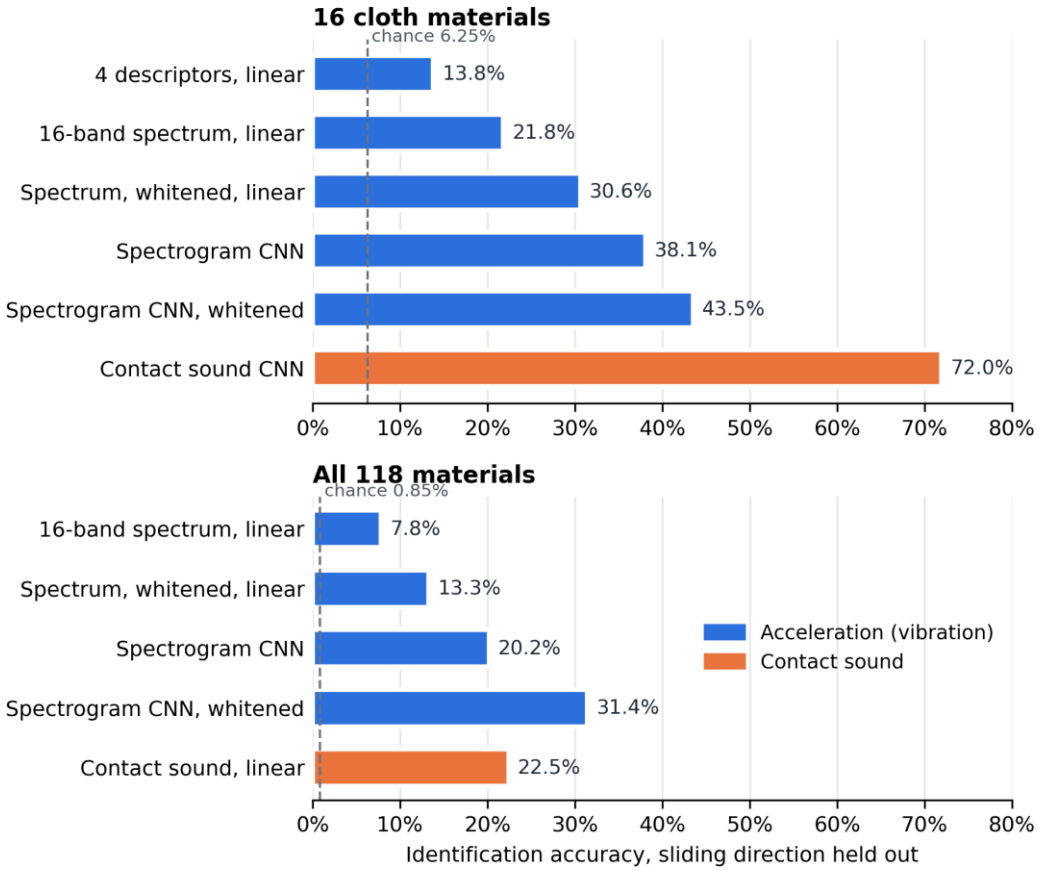


Fig. 4.7 Identification accuracy for every input and model tried in Phase I

Reporting failures matters as much as successes: the O1 gate, the weave group test, analysis N and half of C did not go as predicted, and they are reported as they came out, without changing thresholds afterwards.

4.5 Challenges and Decisions

- The descriptor gate failed. Following the registered plan, this is reported as a result, not tuned away; the richer spectrum and the CNN show where the information goes.
- Several standard measures misbehaved on real data and were replaced: spectral flatness was about zero for every fabric (replaced by spectral entropy), eta-squared is biased at small samples (replaced by omega-squared), and the envelope period was three times too long until the envelope was decimated.
- Coarse fabrics at slow speeds vibrate below the 20 Hz band; such recordings are flagged and excluded, not averaged in.
- An early velocity result was described as pre-registered although its prediction was committed after the run. It has been relabelled as a held-out replication; the numbers are unchanged.

- Some early results were produced from uncommitted code. All eight that this report quotes were re-run from committed code and reproduced exactly; every run now records a hash of its source.
- GPU time is limited, so every step that runs on a CPU was kept off the GPU; only CNN training used it.
- The product-photo set needs two independent human labellers, and the user study needs ethics clearance; both are planned for Phase II, and the main results do not depend on them.

CHAPTER 5
EXPECTED OUTCOMES AND PHASE–II PLAN

5.1 Expected Outcomes Linked to Objectives

Table 5.1 Expected outcomes

Objective	Expected outcome
O1	Published negative result: compact descriptors do not separate fabrics, with the variance decomposition explaining why
O2	The scan-to-photo accuracy gap per textile class on at least 50 labelled product photographs (headline result)
O3	Velocity findings for all 118 materials and the cloth-specific shift; retrieval reported as a consequence of O1
O4	Working Android web interface, with end-to-end latency measured on at least 3 phone models
O5	Ratings of image–vibration match against mismatched controls, if ethics clearance arrives in time

5.2 Project Plan, Milestones and Timeline

Table 5.2 Phase II timeline (16 weeks)

Weeks	Track A – results and paper	Track B – user study
1	Provenance fixes (done), phone vibration test	Submit ethics application
2–3	Collect and label the product-photo set; run transfer	Prepare the rating instrument with real images
3–6	Fix the draft's defects; write the paper	Wait for approval; pilot (about 10 people)
7–8	Internal review; submit	Main study (40 recruited) if approved
9–12	Revisions	Analysis
13–16	Second submission if the study is complete	—

5.3 Work Planned for Phase II

- Run the 30-minute phone test to check whether the rendered patterns can be told apart on real Android phones; its outcome decides the study design.
- Collect at least 50 product photographs covering the textile classes, have two members label them independently, report their agreement, and measure the per-class accuracy gap.
- Connect the real O3 retrieval to the web interface and measure latency on at least three Android models.

- Submit the ethics application, run the pilot and the main study, and analyse the ratings.
- Write and submit the paper to a suitable venue (for example IEEE INDICON or a World Haptics work-in-progress track), after checking its indexing and deadline.

5.4 Conclusions from Phase I

Phase I built and tested the full pipeline on public data and produced clear, reproducible findings. Compact four-number descriptors cannot tell 16 fabrics apart, though the underlying signal carries more identity than they keep. A classifier trained on photographs does not recognise laboratory scans of fabric, and the reverse also fails, so the scan-to-photo gap is real and large. On this dataset the vibration's frequency does not follow sliding speed for any material, but cloth alone shows a speed-dependent shift once the shared response is removed. Phase II will measure the gap on labelled product photographs, complete the interface on real phones, run the user study if approved, and publish the results.

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APPENDICES

Appendix A lists the contribution of each team member. Appendix B maps the project to the Sustainable Development Goals. Appendix C gives the core descriptor-extraction code and Appendix D lists the saved run behind each result. The UML diagram sources for StarUML are in the uml/ folder supplied with this report, and the run identifiers behind every number are in docs/RESULTS.md of the project repository.

APPENDIX A: CONTRIBUTION OF TEAM MEMBERS

Fill in each member's name and registration number. The roles below follow the objective ownership agreed at the start of the project; please confirm them with all members.

Table A.1 Contribution of team members

Member (Name, Regd. No.)	Role	Main responsibilities
Member 1	O1 lead	Haptic descriptors, separability gate, statistics
Member 2	O2 and O3 lead	Image classifiers, scan-to-photo transfer, association and velocity analysis
Member 3	O4 lead	Web interface, vibration rendering, device testing
Member 4	O5 lead	User study design, ethics application, rating instrument

All members took part in planning, reviewing results and preparing this report.

APPENDIX B: RELEVANCE TO SUSTAINABLE DEVELOPMENT GOALS (SDGs)

SDG 12 – Responsible Consumption and Production. Letting shoppers judge a fabric's feel before buying could reduce returns of clothing bought online and the transport and waste they cause. The project does not measure this effect; it is a possible benefit of the application.

SDG 9 – Industry, Innovation and Infrastructure. The project shows how public research data and existing phone hardware can add a new sense to online shopping without new devices.

APPENDIX C: DESCRIPTOR EXTRACTION CODE

The core of Objective 1 is the function below, from haptic/descriptor.py in the project repository. It turns one acceleration recording into the four descriptors (and auxiliary features) used throughout Chapter 4.

```
def extract(accel: np.ndarray, fs: float, *,
            material_id: int = -1,
            velocity_mm_s: float = float("nan"),
            force_n: float = float("nan")) -> Descriptor:
    """Run the full pipeline on one recording.

    Parameters
    ----------
    accel : (N,) or (N, 3) acceleration samples
    fs    : sampling rate in Hz (Cluster acceleration is ~6000)
    """
    x = to_magnitude(accel)
    if len(x) < 64:
        raise ValueError(f"recording too short: {len(x)} samples")

    x = x - np.mean(x)                # remove DC / gravity component
    xb = bandpass(x, fs)
    f, p = _welch(xb, fs)

    centroid = spectral_centroid(f, p)
    env = envelope(xb, fs)

    return Descriptor(
        material_id=material_id,
        velocity_mm_s=velocity_mm_s,
        force_n=force_n,
        frequency_hz=centroid,
        intensity=rms(xb),
        pattern=spectral_entropy(f, p),
        duration_ms=envelope_period_ms(env, fs),
        spectral_rolloff_hz=spectral_rolloff(f, p),
        spectral_bandwidth_hz=spectral_bandwidth(f, p, centroid),
        zero_crossing_rate=zero_crossing_rate(xb, fs),
        env_depth=envelope_depth(env),
        flatness_db=spectral_flatness_db(f, p),
        crest=spectral_crest(f, p),
    )
```

APPENDIX D: SOURCE OF EVERY RESULT

Every number in Chapter 4 comes from a saved run in the repository's results/ folder. Each run stores its configuration, metrics, the Git commit and a hash of the source code. Table D.1 lists the run behind each result.

Table D.1 Saved runs behind the reported results

Result	Run name	Run ID
O1 gate (Table 4.6)	o1_gate	20261108T113942_488140Z
Variance shares, group test, identification (Figs. 4.2, 4.3)	o1_real_analysis	20261108T113952_427442Z
Spectrogram CNN baseline, 16 fabrics	o1_learned_baseline	20261106T214141_196418Z
MINC-2500 and DTD classifiers (Fig. 4.4)	results/o2/	minc2500_fold1_seed0, dtd_fold1–10_seed0
Scan crops and field of view (Fig. 4.5)	o2_cluster_scans / o2_cluster_scan_scale	20261108T133149_279712Z / 20261108T113824_998654Z
Retrieval and velocity, cloth	o3_real	20261108T114001_519385Z
On/off rendering fidelity	o4_fidelity_real	20261108T114014_304295Z
Velocity, all 118 materials (Fig. 4.6)	velocity_all_materials	20261108T114149_659895Z
Held-out velocity replication	velocity_confirm_cloth	20261108T114348_515364Z
A. Looks predict feel	looks_feel	20261107T082554_607613Z
B. Sound vs acceleration, 16 fabrics	audio_vs_accel	20261107T110948_431117Z
C. Unseen velocity	velocity_warp	20261107T111741_349222Z
D. 118-material benchmark	benchmark118_linear / benchmark118_cnn	20261107T103411_132525Z / 20261107T125519_960150Z
E. Direction structure	direction_structure	20261107T140954_426324Z
F. On/off channel	onoff_channel	20261107T142357_330804Z
G. Scan to product photo	photo_transfer	20261107T144009_695846Z
H. Force transfer	force_transfer	20261107T140925_493108Z
I. Whitening, linear	rig_whitening	20261107T152134_578254Z
J. Strict transfer and CLIP	photo_transfer_strict	20261107T154450_969423Z
K. Repeatability	reliability	20261107T152135_235452Z
L. Image to category to descriptor	image_to_descriptor	20261107T153643_203552Z
M. Sound, 118 materials	audio118	20261107T171843_782790Z
N. Whitening and the gate	gate_whitened	20261107T155735_142723Z
O. Velocity invariance	velocity_invariance	20261107T155735_373772Z
P. Whitening, CNN	whitened_cnn	20261107T194114_063058Z